

Molecular Phenotyping of Sepsis and Differential Response to Fluid Resuscitation

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Abstract

Rationale: Biological heterogeneity in critical illness syndromes, such as sepsis and acute kidney injury (AKI), has hindered development of effective therapies. In sepsis-associated AKI, two molecular subphenotypes (SP1 and SP2) have been identified with differing characteristics, outcomes, and response to vasopressor treatment. It is unknown if these subphenotypes extend to all patients with sepsis and whether they respond differently to fluid resuscitation strategy.

Methods: Patients enrolled in the CLOVERS clinical trial with plasma collected at study enrollment were classified into two subphenotypes using a previously validated parsimonious prediction model that included angiopoietin-1 and -2 and soluble tumor necrosis factor receptor-1. Kaplan-Meier estimates were used to test differences in outcomes and subphenotype by treatment interaction.

Measurements and Main Results: Among 1,289 patients, we identified 1,016 as SP1 and 273 as SP2. The risk of poor clinical outcomes was greater in SP2 than in SP1 independent of demographics, comorbidities, and illness severity scores. Patients with SP2, characterized by more severe endothelial injury and inflammation, had higher 28-day mortality with a liberal versus restrictive fluid strategy (41% vs. 27%), whereas patients with SP1 had no difference (9% vs. 9%) (*P* value for interaction = 0.02). Furthermore, SP2 had fewer days free from ventilation, renal replacement therapy, and vasopressors with a liberal compared with a restrictive resuscitation strategy.

Conclusions: Molecular subphenotypes previously identified in AKI are also identifiable in sepsis and respond differently to fluid resuscitation strategy. Future prospective identification of these subphenotypes could inform a precision-guided therapeutic approach in sepsis.

Keywords: sepsis; acute kidney injury; volume; vasopressors; endothelial dysfunction

Sepsis is a life-threatening dysregulated immune response to infection frequently associated with hypotension and multisystem organ dysfunction (1). In critically ill

populations, acute kidney injury (AKI) is the most common form of organ failure in sepsis, developing in as many as 60% of septic patients with hypotension (2, 3).

However, sepsis and AKI are common critical illness syndromes with significant heterogeneity, which may contribute to difficulties in identification of therapeutic

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This article has a related editorial.

A data supplement for this article is available via the Supplements tab at the top of the online article.

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At a Glance Commentary

Scientific Knowledge on the

Subject: Biological heterogeneity in critical illness syndromes, such as sepsis and acute kidney injury (AKI), has hindered the development of effective therapies. It is unknown if previously identified subphenotypes in AKI extend to all patients with sepsis and whether they respond differently to fluid resuscitation strategy.

What This Study Adds to the

Field: Molecular subphenotypes previously identified in AKI are also identifiable in sepsis and respond differently to fluid resuscitation strategy. Future prospective identification of these subphenotypes could inform a precision-guided therapeutic approach in sepsis.

options for prevention or treatment (4, 5). Traditional models of therapies in sepsis have been unsuccessful, focusing on developing therapies that could be provided to broad populations. In contrast, identification of biologically distinct host-response subphenotypes and testing tailored therapies that target distinct disease pathways may prove successful (6).

We previously applied an unsupervised clustering methodology (latent class analysis) to 29 clinical and molecular variables in a population of patients with AKI predominantly caused by sepsis who were admitted to the ICU (7). This data-driven approach identified two subphenotypes (SP1 and SP2) with different molecular characteristics, hospital outcomes, and responses to early addition of vasopressin therapy for septic shock. We also developed and validated a parsimonious three-variable model that included plasma biomarkers of endothelial function (angiopoietin-1 [Ang-1] and angiopoietin-2 [Ang-2]) and inflammation (soluble tumor necrosis factor receptor-1 [sTNFR-1]). Ang-1 and -2 are vascular endothelial growth factors that have opposing activities. Ang-1 is produced by platelets and pericytes, is an agonist for the Tie-2 receptor, and is protective by stabilizing the endothelium, promoting vessel maturation and preventing microcirculatory capillary leakage (8). In contrast, Ang-2

typically acts as an antagonist for binding to the Tie-2 receptor and promotes endothelial dysfunction in the form of vessel destabilization, endothelial cell apoptosis, and capillary leak (9–11). Independently, sTNFR-1 functions in the early innate inflammatory response in sepsis and has been associated with AKI and mortality (12, 13). The combination of these host-response biomarkers identifies a biologically distinct subphenotype in patients with AKI.

Next steps to translate these findings to clinical medicine include 1) determining whether these subphenotypes are generalizable to all patients with sepsis-induced hypotension to speed time to subphenotype recognition; 2) determining whether these subphenotypes can be identified in the emergency room, an earlier and critical time to implement therapeutic strategies; and 3) testing whether these subphenotypes respond differently to volume of intravenous fluids, a therapy that may have a differential response based on a patient's susceptibility to capillary leak. In addition, alternative subphenotypes have been identified in sepsis using either clinical variables (14), gene expression (15), or a combination of clinical and molecular variables (16). We cross-tabulated our subphenotypes with previously established hypo- and hyperinflammatory acute respiratory distress syndrome (ARDS)/sepsis subphenotypes to identify similarities and differences. To address the proposed aims, we used the CLOVERS (Crystalloid Liberal or Vasopressors Early Resuscitation in Sepsis) clinical trial, which randomized patients with sepsis-induced hypotension to an initial restrictive or liberal fluid resuscitation strategy (17). The CLOVERS trial demonstrated no differences in clinical outcomes between treatment arms or in the subgroups with AKI or end-stage renal disease (ESRD) or stratifying patients by Sequential Organ Failure Assessment scores (17, 18). We hypothesized that the liberal fluid resuscitation strategy would be harmful to SP2, a subtype with elevated markers of endothelial injury and inflammation, compared with SP1. Some of the results of these studies were previously reported in the form of an abstract (19).

Methods

Study Design and Oversight

We conducted a secondary analysis of data from CLOVERS, a multicenter, prospective, phase III, randomized, nonblinded trial.

Patients enrolled in CLOVERS were allocated to either a restrictive or liberal fluid resuscitation strategy in the setting of sepsis. All patients were randomized within 4 hours of meeting inclusion criteria, and the duration of protocol for restrictive or liberal fluid resuscitation was 24 hours. Patients randomized to a restrictive fluid protocol had vasopressors prioritized, with “rescue fluids” being permitted for prespecified indications that suggested severe intravascular volume depletion. The liberal fluid protocol consisted of a recommended initial 2-L intravenous infusion of isotonic crystalloid, followed by fluid boluses administered on the basis of clinical triggers with “rescue vasopressors” permitted for prespecified indications. A more detailed description of the methods can be found in the supplement of the primary publication (17).

Subphenotype Identification

Methods for measurement of Ang-1, Ang-2, sTNFR-1, and IL-6 can be found in the data supplement. We used plasma biomarker concentrations of Ang-1, Ang-2, and sTNFR-1 to identify two subphenotypes (SP1 and SP2). Since the publication of our prior works, the sTNFR-1 assay changed. In analyses provided in the data supplement, we found the correlation of these two assays to be high (Pearson's correlation, $r = 0.94$), and we created a calibration equation (see Figure E2 in the data supplement). Participants with a probability of SP2 > 0.5 were then classified as having the SP2 subphenotype. We also used a validated three-variable model that included IL-6, sTNFR-1, and serum bicarbonate to classify ARDS/sepsis subphenotypes (20).

AKI Definition

AKI was defined using the KDIGO (Kidney Disease: Improving Global Outcomes) criteria as an increase in serum creatinine at randomization of $\geq 50\%$ or ≥ 0.3 mg/dl above a baseline serum creatinine. Urine output was not used to define AKI because of unavailability. The lowest outpatient or inpatient serum creatinine value within the last year before the index hospitalization was used for the baseline when available ($n = 909$). If no prehospitalization serum creatinine was available and there was no known history of chronic kidney disease (CKD), then we imputed a baseline serum creatinine on the basis of an estimated glomerular filtration rate of 90 ml/min/1.73 m² ($n = 517$) using the 2021 creatinine-based CKD-EPI (Chronic Kidney Disease

Epidemiology Collaboration) equation (7). If no prehospitalization serum creatinine was available and there was a known history of CKD, then we used the lowest serum creatinine value during the index hospitalization as the baseline ($n = 29$). Patients were subsequently staged according to the serum creatinine at randomization using the KDIGO guidelines. Seventy-five patients were undergoing maintenance hemodialysis before study enrollment (for ESRD).

Outcome Measures

The primary outcome for this analysis was 28-day and 90-day mortality. Additional outcomes included number of days free from ventilator use, number of days free from renal replacement therapy, and number of days free from vasopressor use within 28 days.

Statistical Analysis

To test for heterogeneity of treatment effect, we completed an intention-to-treat comparison of the treatment effect of restrictive or liberal fluid resuscitation strategy on 28-day and 90-day mortality by subphenotype. For this analysis, we used Kaplan-Meier 28-day and 90-day mortality point estimates. Patients who were lost to follow-up before Day 90 were censored at the time they were last confirmed to be alive. We compared 28-day and 90-day mortality point estimates and evaluated the evidence for treatment by subphenotype interaction using a z test with Greenwood's SE and a 95% Wald confidence interval (CI) for the difference in mortality. For analysis of secondary outcomes, we used linear regression with robust SEs to estimate the difference in number of days free from organ support between the two treatment groups.

Finally, we compared the risk of death by subphenotype over 28 and 90 days of follow-up using Cox regression, adjusting for potential confounders decided *a priori*. For all analyses of 28-day mortality, the proportional hazards assumption was not rejected ($P > 0.05$ for all models). In contrast, for all analyses of 90-day mortality, the proportional hazards assumption for all models was rejected ($P < 0.01$ for all). Inspection of the Schoenfeld residuals over 90 days of follow-up indicated attenuation of the treatment effect during the latter stages of follow-up. Thus, the estimated hazard ratios (HRs) for the outcome of 90-day mortality represent the time-averaged association over follow-up (21). All analyses were conducted

using the R 4.2.3 software environment (R Foundation for Statistical Computing). Two-sided P values < 0.05 were taken as evidence of statistical significance.

Results

Baseline Characteristics of Participants by Sepsis-associated AKI Subphenotype

Among 1,563 participants enrolled in CLOVERS, 259 were excluded from this analysis because of unavailable plasma and 15 were excluded because of no serum creatinine measured at randomization (Figure E1). Comparison of patients included and excluded in this study demonstrated similar baseline characteristics between patient populations (Table E1). Among 1,289 patients with available biospecimens, we applied a previously published subphenotype classification model that incorporated plasma Ang-1, Ang-2, and sTNFR-1 measurements. We identified 1,016 (79%) patients as SP1 and 273 (21%) as SP2 (Table 1). Approximately half of patients in each subphenotype were in each treatment arm. Patients with SP2 had greater rates of congestive heart failure and CKD and more severe KDIGO stage of AKI at study enrollment than SP1.

Clinical Outcomes by Subphenotype

To evaluate the association of subphenotypes with clinical outcomes, patients with SP1 and SP2 were directly compared (Table E2). In models adjusting for demographics, comorbidities, KDIGO stage of AKI, baseline Sequential Organ Failure Assessment scores, and baseline norepinephrine dose, patients with SP2 had greater risk of 28-day (HR, 2.36; 95% CI, 1.68, 3.33) and 90-day mortality (HR, 1.93; 95% CI, 1.45, 2.56) than SP1. Patients with SP2 were also more likely than those with SP1 to have fewer days free from ventilator use, renal replacement therapy, or vasopressor use at 28 days (Table E3).

Difference in Volume of Fluid and Timing of Vasopressors by CLOVERS Randomization

In the 24-hour protocol after randomization, patients with SP1 randomized to the liberal fluid group received, on average, 2 L more intravenous fluids than patients with SP1 randomized to the restrictive fluid group ($3,684 \pm 1,644$ ml vs. $1,684 \pm 1,657$ ml) (Table E4). Similarly, patients with SP2

randomized to the liberal fluid group received, on average, 4,091 ($\pm 1,793$) ml compared with 2,244 ($\pm 2,305$) ml in the restrictive fluid group. Patients randomized to the restrictive fluid group received vasopressors more often than patients in the liberal fluid group across both subphenotypes.

Interaction between Subphenotype and Resuscitation Strategy

Next, we determined whether subphenotypes identified using biospecimens collected before randomization respond differently to the fluid resuscitation strategy. In SP1, a similar proportion of patients died at 28 days in the liberal and restrictive fluid groups (9% vs. 9%, respectively) (Figure 1). In SP2, 28-day mortality was greater in the liberal fluid group than in the restrictive fluid group (41% vs. 27%, respectively), and the difference in treatment effects between SP1 and SP2 was 14% (95% CI, 2%, 25%; P value for interaction = 0.02). In addition, 90-day mortality was also greater with the liberal than with the restrictive fluid group in SP2 (48% vs. 36%), whereas SP1 demonstrated no difference (15% vs. 17%) (P value for interaction = 0.03) (Table 2). Next, we evaluated the continuous probability of belonging to SP2 and the interaction of a liberal versus a restrictive fluid strategy. We observed a significant interaction between the probability of subphenotype and fluid resuscitation strategy and 28-day mortality ($P = 0.008$). Because patients had a greater certainty of the SP2 phenotype (i.e., higher probability of SP2 based on the three-biomarker model), we observed a larger difference in 28-day mortality between the liberal and restrictive fluid groups.

To understand the potential causes for differences in 28- and 90-day mortality in SP2, we evaluated secondary outcomes, such as number of days free from renal replacement therapy, vasopressors, and mechanical ventilation at 28 days. We found no differences for the secondary outcomes between the restrictive and liberal fluid groups in patients with SP1. In contrast, patients in SP2 had improved outcomes in the restrictive fluid group compared with the liberal fluid group. Patients with SP2 had, on average, 3.2 more days free from vasopressor use, 2.7 more days free from renal replacement therapy use, and 3.4 more days free from ventilator use at 28 days in the restrictive compared with the liberal fluid

Table 1. Baseline Characteristics, Stratified by Subphenotype

	SP1 (n = 1,016)		SP2 (n = 273)	
	Liberal (n = 510)	Restrictive (n = 506)	Liberal (n = 137)	Restrictive (n = 136)
Age, yr, average $\pm$ SD	58.8 $\pm$ 16.2	59.4 $\pm$ 15.5	64.8 $\pm$ 13.8	61.3 $\pm$ 16.5
Male, n (%)	265 (52%)	256 (51%)	86 (63%)	75 (55%)
Race, n (%)				
White	384 (75%)	355 (70%)	94 (69%)	87 (64%)
Black	62 (12%)	85 (17%)	28 (20%)	31 (23%)
Asian	17 (3%)	16 (3%)	4 (3%)	4 (3%)
Other	5 (1%)	5 (1%)	0 (0%)	3 (2%)
Not reported	42 (8%)	45 (9%)	11 (8%)	11 (8%)
Ethnicity, n (%)				
Hispanic/Latino	72 (14%)	84 (17%)	18 (13%)	16 (12%)
Not Hispanic/Latino	422 (83%)	400 (79%)	115 (84%)	116 (85%)
Not reported	16 (3%)	22 (4%)	4 (3%)	4 (3%)
Randomization location, n (%)				
ED	473 (93%)	477 (94%)	115 (84%)	122 (90%)
ICU	32 (6%)	23 (5%)	20 (15%)	14 (10%)
Other	5 (1%)	6 (1%)	2 (1%)	0 (0%)
Invasive mechanical ventilation, n (%)	32 (6%)	28 (6%)	10 (7%)	12 (9%)
Source of infection, n (%)				
Pneumonia	153 (30%)	152 (30%)	31 (23%)	34 (25%)
Urinary tract infection	136 (27%)	110 (22%)	37 (27%)	25 (18%)
Intra-abdominal infection	46 (9%)	46 (9%)	21 (15%)	16 (12%)
Skin or soft tissue infection	47 (9%)	62 (12%)	13 (9%)	10 (7%)
Other or unknown	128 (25%)	136 (27%)	35 (26%)	51 (38%)
Weight, kg, average $\pm$ SD	77.7 $\pm$ 26.1	78.6 $\pm$ 25.9	80.0 $\pm$ 21.8	82.5 $\pm$ 26.1
BMI, kg/m ² , average $\pm$ SD	27.2 $\pm$ 8.3	27.4 $\pm$ 7.8	27.3 $\pm$ 7.7	29.0 $\pm$ 8.9
Diabetes mellitus, n (%)	144 (28%)	137 (27%)	46 (34%)	56 (41%)
Chronic heart failure, n (%)	46 (9%)	59 (12%)	22 (16%)	28 (21%)
History of CKD, n (%)	30 (6%)	28 (6%)	41 (30%)	40 (29%)
Baseline serum creatinine, mg/dl, average $\pm$ SD*	0.8 $\pm$ 0.4	0.8 $\pm$ 0.4	1.1 $\pm$ 1.2	1.0 $\pm$ 0.9
SOFA score, average $\pm$ SD	2.9 $\pm$ 2.3	2.8 $\pm$ 2.3	5.8 $\pm$ 2.9	5.5 $\pm$ 3.1
Volume of fluid administered before randomization, ml, median (IQR)	2,050 (1,411–2,500)	2,055 (1,500–2,500)	2,050 (1,350–2,500)	2,100 (1,500–2,433)
KDIGO stage of AKI at study enrollment, n (%)				
No AKI	205 (40%)	206 (41%)	14 (10%)	18 (13%)
Stage 1 AKI	122 (24%)	144 (28%)	25 (18%)	20 (15%)
Stage 2 AKI	106 (21%)	85 (17%)	23 (17%)	29 (21%)
Stage 3 AKI	69 (14%)	63 (12%)	46 (34%)	39 (29%)
End-stage renal disease, n (%)	8 (2%)	8 (2%)	29 (21%)	30 (22%)
Biomarker concentrations, median (IQR)				
sTNFR-1	4,341 (2,489–6,695)	4,407 (2,806–7,334)	17,996 (10,762–26,279)	17,738 (11,828–24,952)
Ang-1	5,713 (2,860–10,255)	5,545 (2,999–9,331)	1,548 (595–3,749)	1,912 (905–3,658)
Ang-2	6,880 (4,514–10,767)	7,674 (4,799–12,186)	22,955 (15,080–37,105)	21,956 (16,753–37,689)

Definition of abbreviations: Ang-1 = angiotensin-1; Ang-2 = angiotensin-2; BMI = body mass index; CKD = chronic kidney disease; ED = emergency department; IQR = interquartile range; KDIGO = Kidney Disease: Improving Global Outcomes; SOFA = Sequential Organ Failure Assessment; SP = subphenotype; sTNFR-1 = soluble tumor necrosis factor receptor-1.

Entries are mean (SD) for continuous variables and count (percent) for categorical variables, except as noted.

*Excluding patients with a history of end-stage renal disease.

group (all *P* values for interaction ≤ 0.05) (Table 3).

Interaction between Subphenotype and Resuscitation Strategy in the Subgroup with AKI and ESRD

In sensitivity analyses, we evaluated the treatment interaction of fluid resuscitation and subphenotypes restricted to patients with AKI or ESRD at study enrollment (*n* = 846). This study population may be

particularly prone to complications from a liberal fluid strategy because of systemic inflammation and susceptibility for fluid retention. We found that, consistent with the overall results, SP2 was associated with worse clinical outcomes in the liberal fluid group. We observed a significant interaction between the probability of SP2 and fluid resuscitation strategy and 28-day mortality (*P* = 0.01) (Figure 2). Among 443 patients without AKI or ESRD at study enrollment,

only 32 (7%) patients were classified as SP2, and we identified no heterogeneity of treatment effect in this population without AKI (Table E5).

Cross-Tabulation of AKI/Sepsis and ARDS/Sepsis Subphenotypes

We directly compared the cross-tabulation of SP1 and SP2 and ARDS/sepsis subphenotypes (hypoinflammatory and hyperinflammatory). Cross-tabulation demonstrated mild to

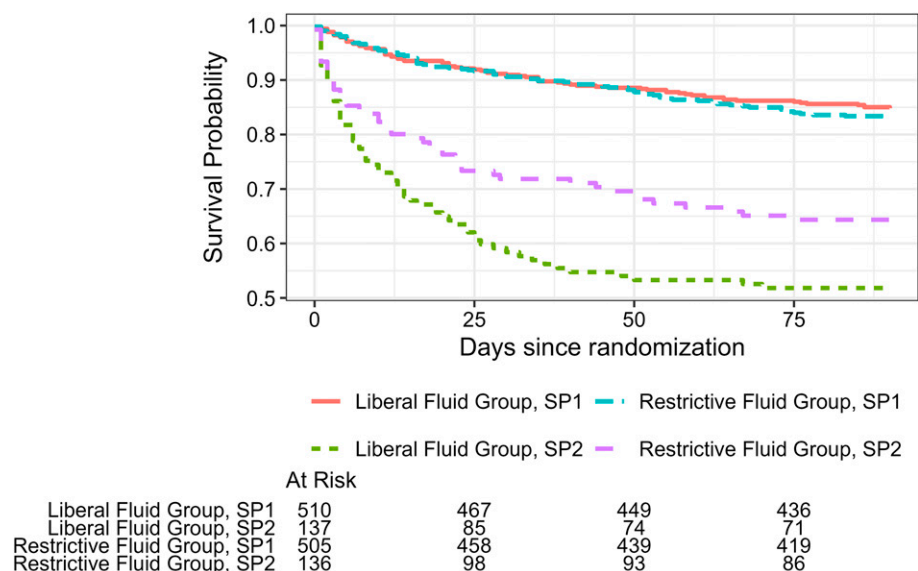


Figure 1. Kaplan-Meier survival curve in sepsis subphenotypes (SPs), stratified by resuscitation treatment group. Curves for each SP are stratified by liberal or restrictive resuscitation strategy. Below the x-axis, patients at risk are provided for study enrollment and Days 25, 50, and 75.

moderate overlap between hypoinflammatory and SP1 and between hyperinflammatory and SP2 but also demonstrated distinct differences (Cohen's kappa for agreement, 0.44) (Table E6). Out of 1,289 patients, 81.4% were concordant (i.e., hypoinflammatory and SP1 or hyperinflammatory and SP2), and 18.6% were discordant. We observed no differential response to a restrictive or liberal fluid resuscitation strategy between ARDS/sepsis subphenotypes for 28-day (P value for interaction = 0.50) or 90-day mortality (P value for interaction = 0.62) (Table E7).

Discussion

Heterogeneity in sepsis and AKI has hindered development of interventions that target an individual patient's pathophysiology (22). We completed a secondary analysis of the CLOVERS trial and identified molecular subphenotypes that

predicted differential treatment response to liberal versus restrictive fluid resuscitation strategies, revealing a treatment effect that was not seen in the original CLOVERS study. A potential biological mechanism for the differential treatment response is that patients with SP2 have greater plasma markers of endothelial dysfunction, and a liberal fluid resuscitation strategy may lead to worsening organ injury (more days receiving vasopressors, mechanical ventilation, and renal replacement therapy) and, in turn, greater 28- and 90-day mortality. Taken together, these data offer a precision medicine treatment strategy in sepsis-induced hypotension.

In sepsis, time to antibiotics and supportive therapy are critical to improve outcomes. Timely identification of subphenotypes is a requisite for a directed therapeutic approach to be successful. However, identification of AKI at hospital presentation requires knowledge of a

patient's baseline kidney function, which often may be missing (23). In this study, we were able to translate previously identified subphenotypes in sepsis-associated AKI to all patients with sepsis-induced hypotension. We found that a three-biomarker model remained effective for subphenotype identification in the emergency department, an earlier and critical window for therapeutic decisions in sepsis treatment. We also found that 93% of patients without AKI upon study enrollment were classified as SP1. In this manner, identification of molecular subphenotypes among all patients with sepsis-induced hypotension may improve generalizability and time to subphenotype identification to facilitate future clinical trials without misclassification of patients with SP2.

SP2 was characterized by markers consistent with endothelial injury (high Ang-2/Ang-1) and inflammation (high sTNFR-1) compared with SP1. In addition,

Table 2. Treatment Effects, by Subphenotype Status in CLOVERS

CLOVERS Population ($n = 1,289$)		No. at Risk (% Who Died), Liberal	No. at Risk (% Who Died), Restrictive	Difference, Restrictive – Liberal (95% CI)	Difference of Difference (95% CI)	P Value for Interaction
28-d mortality	SP1	510 (9%)	506 (9%)	0% (–3%, 4%)	14% (2%, 25%)	0.02
	SP2	137 (41%)	136 (27%)	–13% (–25%, –2%)		
90-d mortality	SP1	510 (15%)	506 (17%)	1% (–3%, 6%)	14% (2%, 26%)	0.03
	SP2	137 (48%)	136 (36%)	–13% (–24%, –1%)		

Definition of abbreviations: CI = confidence interval; SP = subphenotype.

Table 3. Secondary Outcomes, by Subphenotype Status in CLOVERS

	Subphenotype	Mean (SD), Liberal	Mean (SD), Restrictive	Difference (Liberal Is Reference Group) (95% CI)	P Value for Interaction
Number of days free from vasopressor use at 28 d	SP1	23.3 (7.7)	23.3 (7.6)	0 (−0.9, 0.9)	0.006
	SP2	15.0 (12.6)	18.3 (11.4)	3.2 (0.4, 6.1)	
Number of days free from renal replacement therapy at 28 d	SP1	25.4 (8.0)	25.3 (8.1)	−0.1 (−1.1, 0.9)	0.047
	SP2	16.9 (13.6)	19.6 (12.7)	2.7 (−0.8, 6.1)	
Number of days free from ventilator use at 28 d	SP1	24.6 (8.5)	24.7 (8.5)	0.2 (−0.9, 1.2)	0.050
	SP2	15.9 (13.5)	19.3 (12.7)	3.4 (0.3, 6.6)	

Definition of abbreviations: CI = confidence interval; SP = subphenotype.

the SP2 population had high rates of severe AKI, CKD, and ESRD. Because these biomarkers are unlikely to undergo glomerular filtration or dialytic clearance because of their size (approximately >50 kD) (24) and are not appreciably found in the urine of healthy (25) or critically ill patients

(26), and because multiple studies have demonstrated elevations in these biomarkers in patients with kidney injury (27–29), the susceptibility of patients with kidney injury to SP2 may reflect alterations in metabolic processes in parallel to kidney function. A potential mechanism for the differential

response to resuscitation strategy is that patients with more severe endothelial injury and kidney injury are more prone to capillary leak with a liberal fluid resuscitation strategy (30, 31). Supportive of this hypothesis is that the SP2 population had fewer days free from renal replacement therapy, vasopressors, and ventilation with a liberal compared with a restrictive resuscitation strategy. Moreover, in subgroup analyses, we found that heterogeneity in treatment response was most evident in patients with AKI or ESRD upon study enrollment. The combination of oliguria or anuria with a host response of endothelial injury may place a patient at higher risk of capillary leak, interstitial edema, and poor outcomes with a liberal fluid resuscitation strategy.

A reanalysis of the Protocolized Care for Early Septic Shock (ProCESS) (32) trial also demonstrated heterogeneity of treatment effect, with the population with greater inflammation and endothelial injury demonstrating harm with a liberal fluid resuscitation strategy. In the ProCESS trial, investigators identified two subphenotypes, with subphenotype 1 (characterized by greater markers of inflammation) having a greater 60-day inpatient mortality with early goal-directed therapy fluid resuscitation than a usual care fluid strategy. On average, patients randomized to protocol-based early goal-directed therapy received approximately 600 ml of additional fluid compared with the usual care group in the first 6 hours of sepsis resuscitation. Acknowledging the differences in the examined treatments and populations in this prior trial, we discovered that the overall findings are consistent with the proposed mechanism that patients with more endothelial injury and inflammation have worse outcomes with a liberal than with a restrictive fluid resuscitation strategy. In the present reanalysis of the CLOVERS trial,

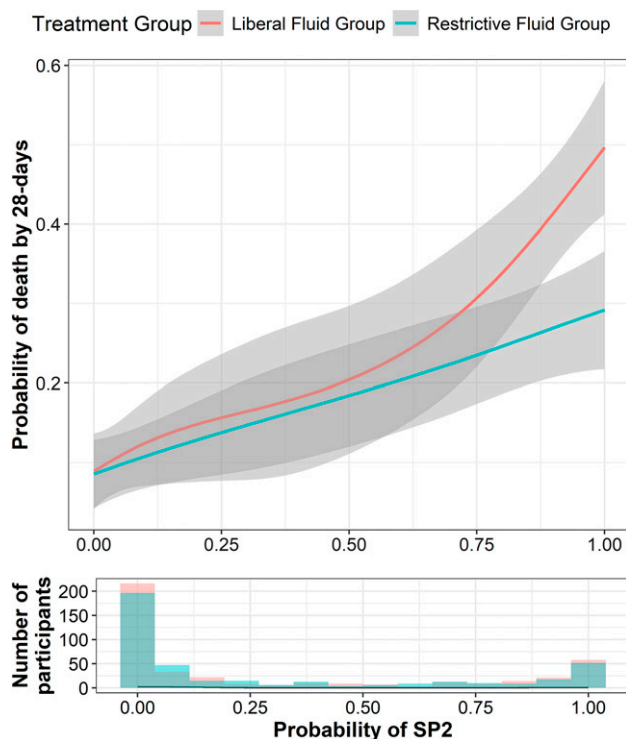


Figure 2. Probability of belonging to subphenotype 2 (SP2) and differential treatment response to a restrictive and liberal fluid resuscitation strategy restricted to patients with acute kidney injury (AKI) and end-stage renal disease upon study enrollment. Upper panel shows probability of belonging to SP2 (x-axis) and the probability of 28-day mortality (y-axis). The lines plot the estimated mortality in either a liberal (red) or restrictive (blue) resuscitation strategy with 95% confidence intervals over a range of probabilities of assignment to SP2. The *P* value tests the null hypothesis of no interaction between treatment group and the continuous probability of SP2 membership and was obtained from a linear regression of binary 28-day mortality on treatment group, continuous probability of SP2 membership, and their interaction. Lower panel provides a density plot that shows the distribution of AKI SP2 probabilities by treatment group.

we demonstrate, in patients with sepsis-induced hypotension, that a parsimonious set of plasma biomarkers can reliably identify patients who differentially benefit from a liberal compared with a restrictive fluid resuscitation strategy.

Previously, multiple sepsis subphenotypes have been identified, and each may be representative of a particular biological mechanism or treatable trait (33). Here, we demonstrate that sepsis subphenotypes defined by differences in endothelial dysfunction predicted response to fluid resuscitation strategy. In contrast, hypoinflammatory or hyperinflammatory sepsis subphenotypes did not demonstrate differences in clinical outcomes on the basis of fluid resuscitation strategy. One explanation for these findings is that parsing sepsis on the basis of endothelial dysfunction may identify populations that differentially benefit from therapies that predominantly target the vasculature, such as vasopressors or resuscitation strategy (7, 16). However, it is important to note that each subphenotype strategy leverages overlapping elements of a dysregulated host response and may be more broadly in common than different. Further research is necessary to compare different phenotyping approaches in critical illness in different clinical populations.

One potential takeaway from this study is that all patients with sepsis-induced hypotension should be treated with a restrictive fluid and early vasopressor resuscitation strategy. However, the addition of early vasopressors for hypotension may also lead to increased resource use without differences in clinical outcomes. In many hospital systems in the United States, initiation of continuous vasopressors necessitates transfer to the ICU. In patients classified as SP1, a liberal fluid strategy led

to fewer patients requiring vasopressors for hypotension (30% vs. 56%) with no difference in 28-day or 90-day mortality. Thus, a liberal fluid strategy in SP1 may not be associated with harm and potentially could decrease use of scarce hospital resources, such as ICU beds. In contrast, a liberal fluid resuscitation strategy in SP2 was associated with harm. These findings highlight that a one-size-fits-all approach may be less effective in patients with sepsis-induced hypotension.

Strengths and Limitations

Our study has several strengths. First, we were able to leverage the randomization in the CLOVERS trial to address limitations of indication bias of fluid administration in observational studies. Second, we directly compared different sepsis subphenotypes to highlight similarities and differences and demonstrate that each subphenotype may represent a treatable trait that requires tailoring a potential therapeutic with the underlying biological mechanism. Third, the CLOVERS trial included patients with ESRD, a group that is often excluded from large randomized clinical trials (34). Although limited by small group size, our data suggest that most patients with ESRD have a subphenotype consistent with SP2 (79%) and may benefit from a restrictive fluid resuscitation strategy.

This analysis should be interpreted in the context of its limitations. First, there was variation in the amount of fluid received in the CLOVERS trial in the restrictive and liberal fluid groups, and overlap of total volumes between groups may dampen a potential therapeutic signal. Second, this was a retrospective study of a completed clinical trial. However, the study design and prediction model for subphenotype identification were defined *a priori*. Third,

a subset of patients did not have available plasma for analysis and were not included. However, comparing patients included in this analysis with patients with missing data did not demonstrate significant differences in baseline characteristics, potentially limiting the concerns for selection bias. Fourth, urine output data before randomization were not available. Approximately 90% of patients were enrolled from the emergency department in a median of 1 hour after meeting trial eligibility. Thus, patients did not have the requisite 6–12 hours or more of monitoring to determine AKI status on the basis of urine output before clinical trial enrollment. Fifth, translating findings to clinical care requires developing point-of-care assays with rapid turnaround time, prospectively identifying subphenotypes, and completing subphenotype-stratified randomized control trials.

Conclusions

In summary, this work posits that early identification of subphenotypes can inform clinical decision making in patients with sepsis-induced hypotension. We observed that SP2 had a greater 28-day and 90-day mortality with a liberal fluid resuscitation strategy. We also observed that SP2 had fewer days free from ventilation, vasopressors, and renal replacement therapy with a liberal fluid resuscitation strategy, suggesting a possible mechanism for the greater mortality. Identification of subphenotypes in patients with sepsis-induced hypotension may facilitate development of targeted treatment interventions in critically ill patients. ■

Author disclosures are available with the text of this article at www.atsjournals.org.

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